

Fundamentals of Solid State Physics

Electronic Band Diagram - Summary

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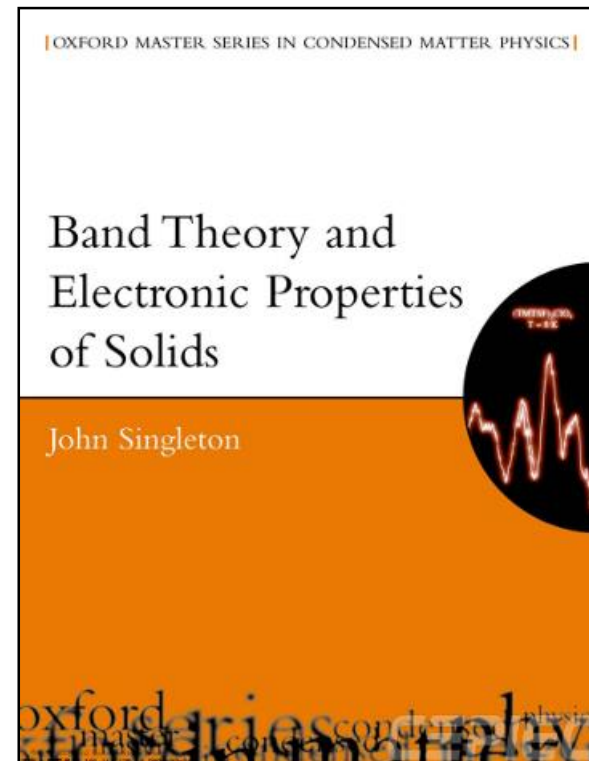
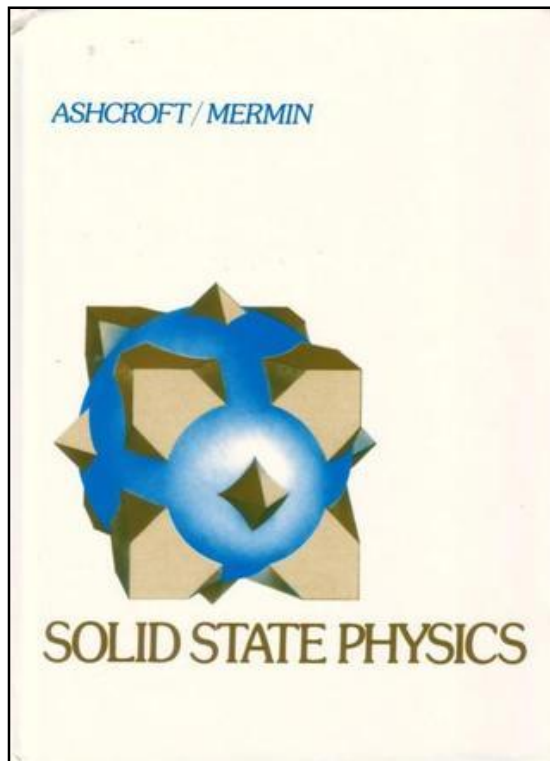


This Class

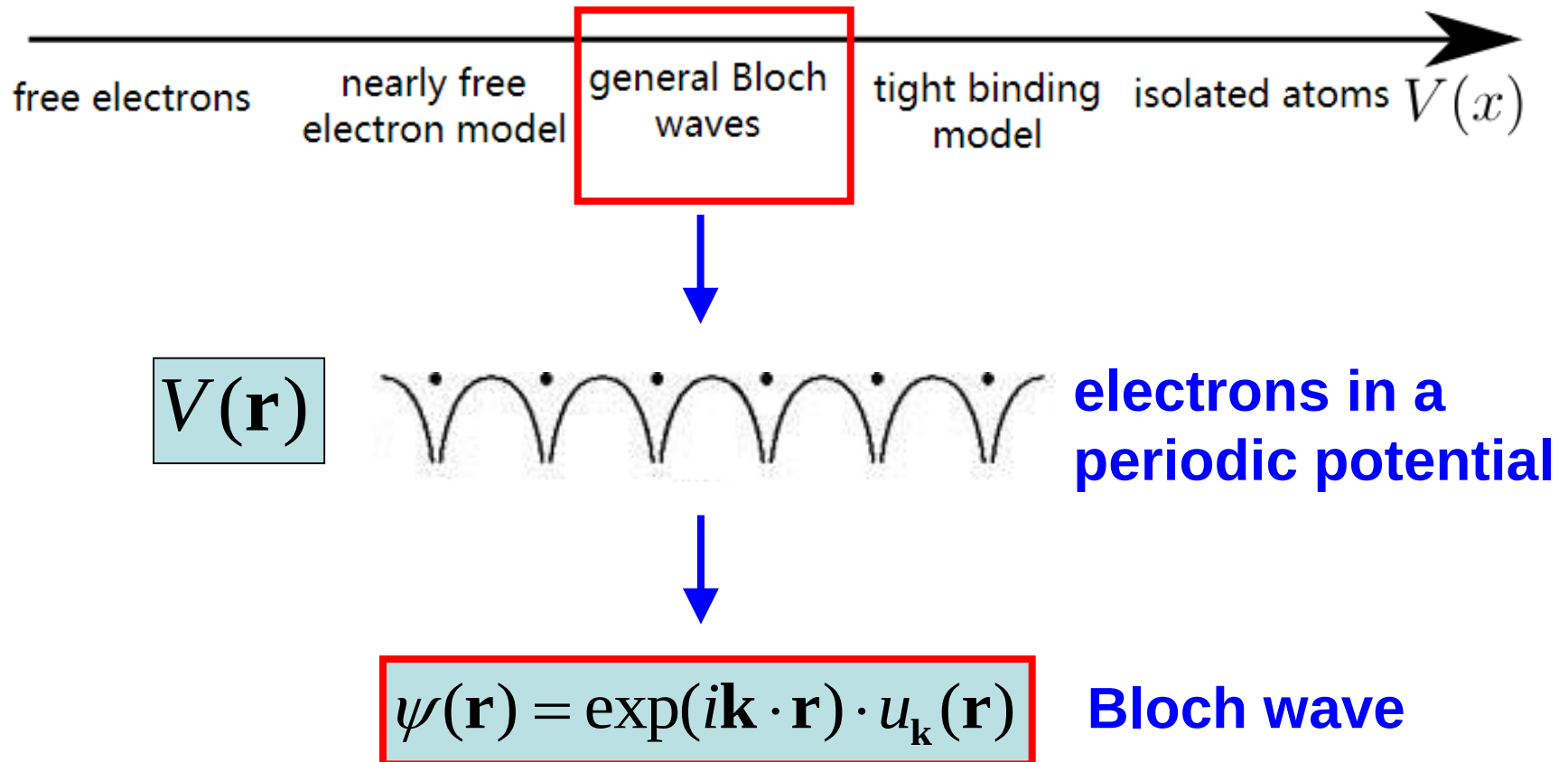
- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Further Reading

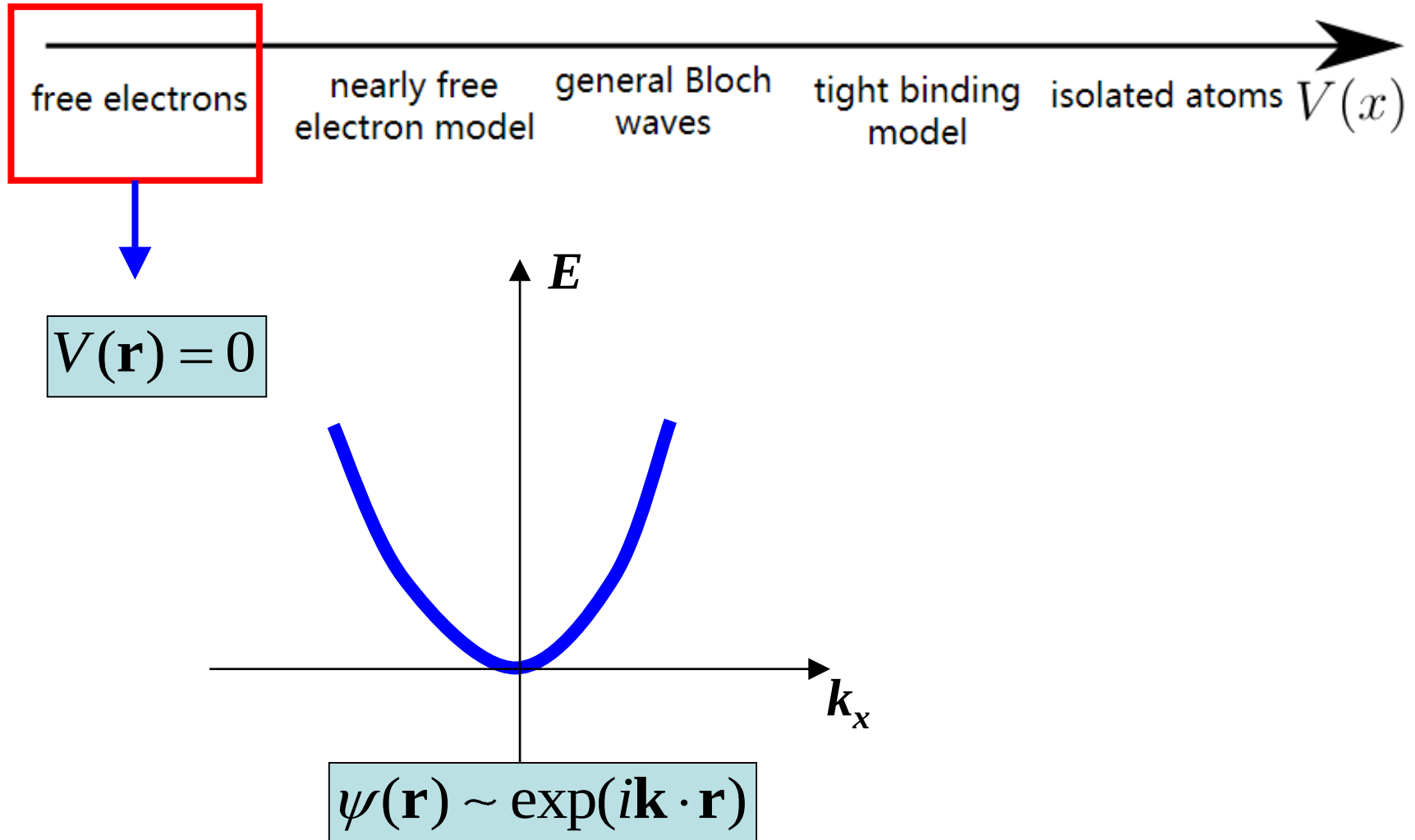
- Singleton, Chapter 5
- Ashcroft & Mermin, Chapter 12, 13



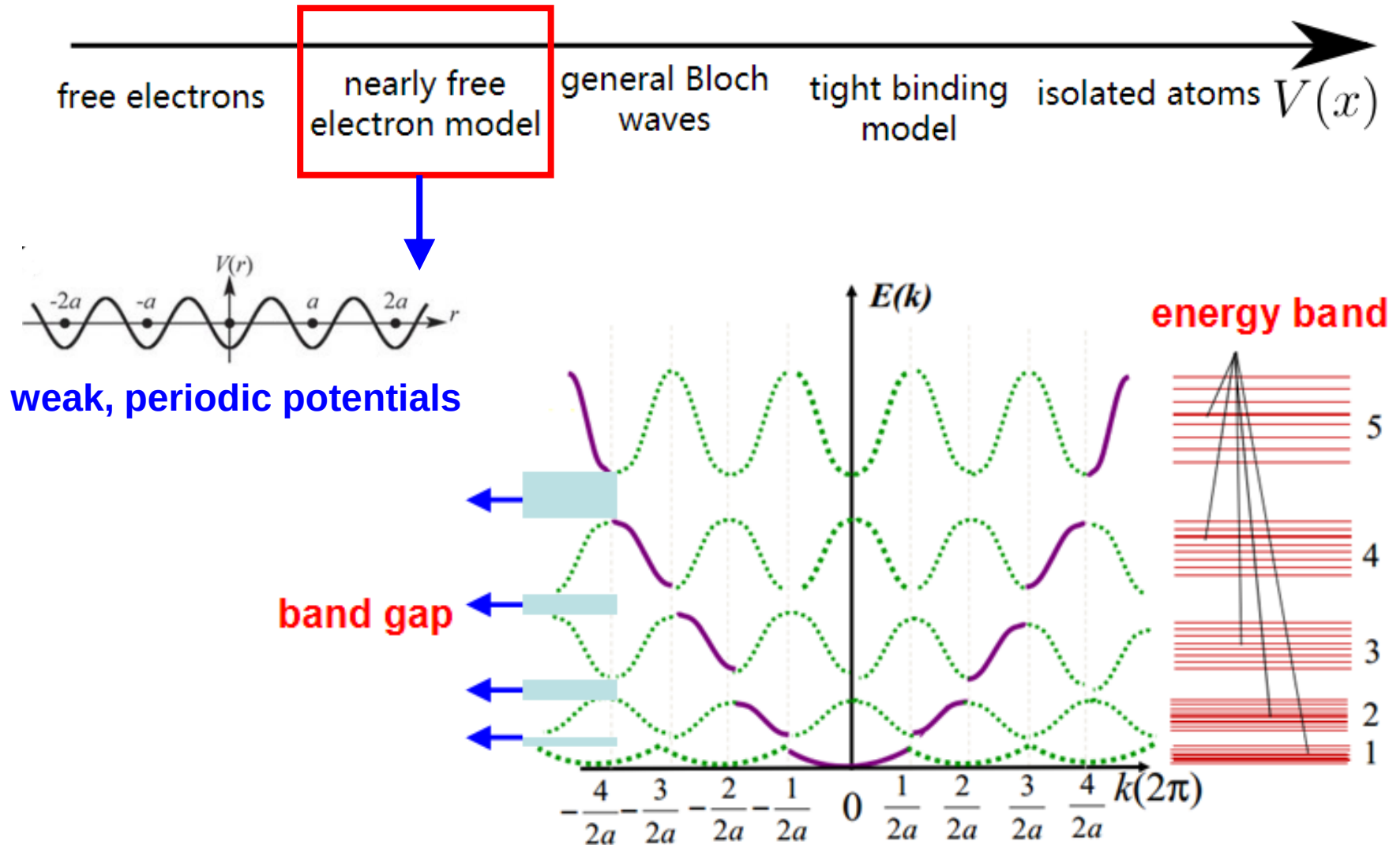
Summary



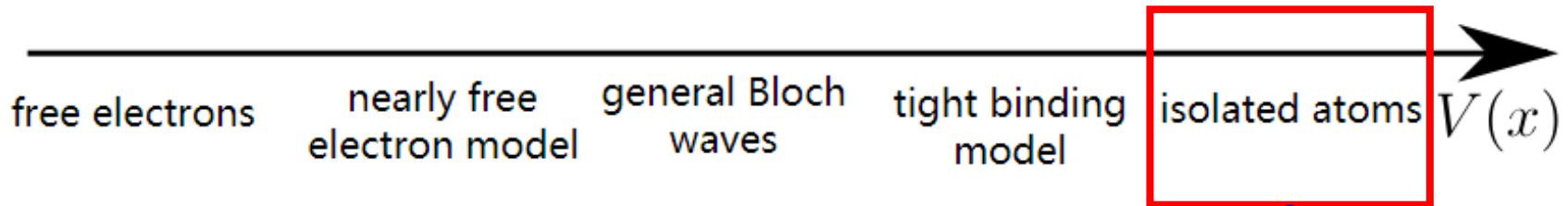
Summary



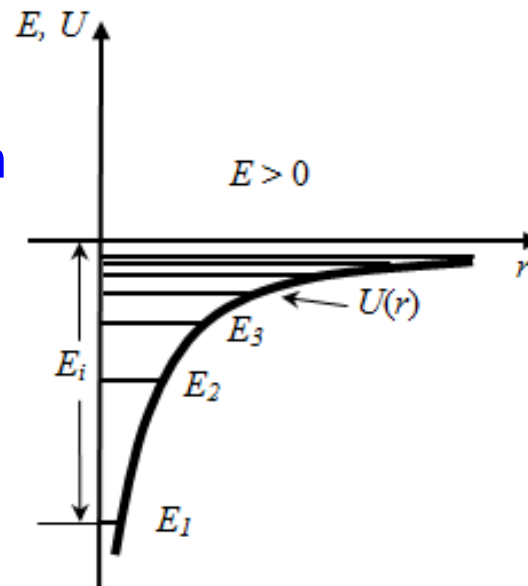
Summary



Summary



hydrogen atom

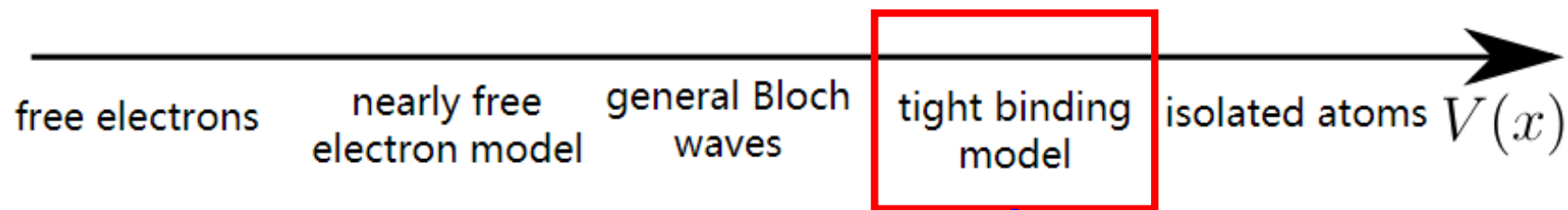


$$V(\mathbf{r}) = -\frac{e^2}{4\pi\epsilon_0} \frac{1}{r}$$

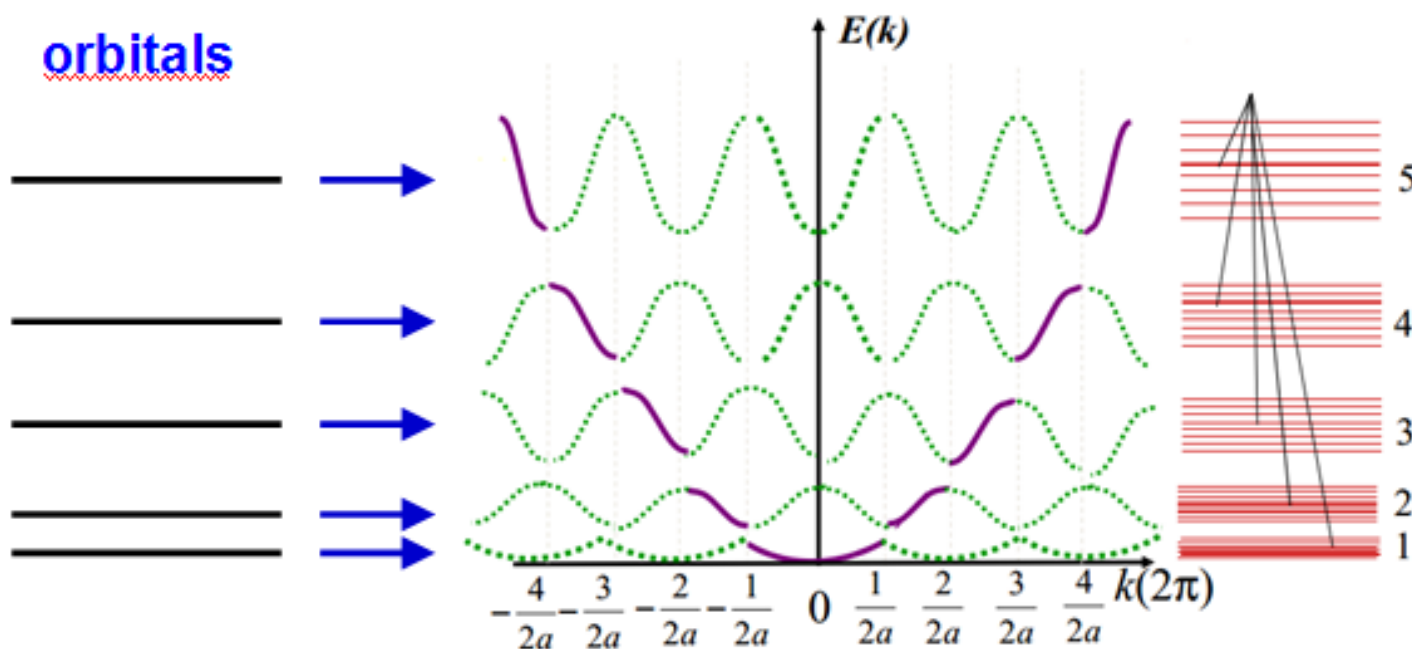
$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

$$\psi(r, \theta, \varphi) = R_{nl}(r) \cdot Y_{lm}(\theta, \varphi)$$

Summary



orbitals



**band structures
and band gaps**

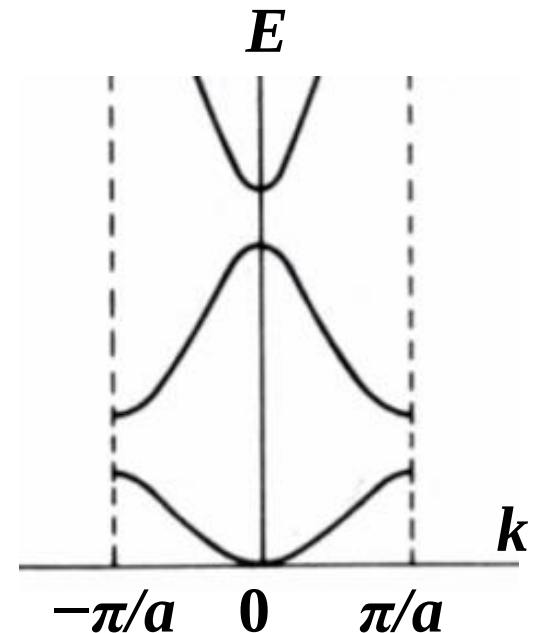
Importance of k

- k is taken from reciprocal space

- ▣ reduced to FBZ

$$k_x = \frac{2\pi n_x}{L_x} \quad n_x = 0, \pm 1, \pm 2, \dots$$

- $p = \hbar k$ is not electron momentum, is **crystal momentum** (take the crystal as a whole)



- group velocity

- ▣ velocity of a wave packet (波包)

$$v_g = \frac{1}{\hbar} \frac{dE(k)}{dk} \quad \text{or} \quad \mathbf{v}_g = \frac{1}{\hbar} \nabla_{\mathbf{k}} E(\mathbf{k})$$

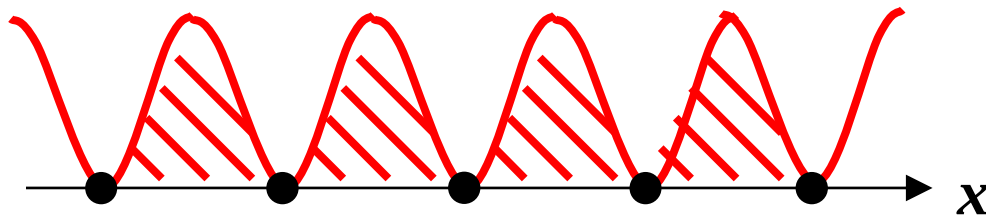
Group Velocity

- At the top and the bottom of the energy bands, $v_g = 0$

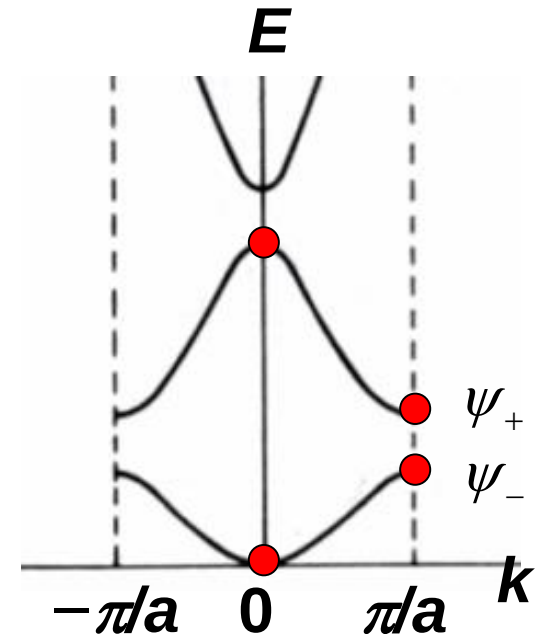
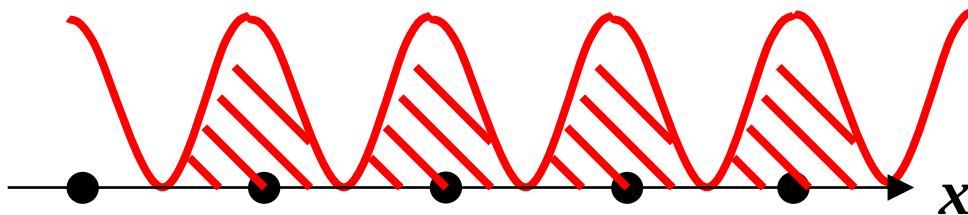
□ standing waves

$$v_g = \frac{1}{\hbar} \frac{dE(k)}{dk}$$

$$|\psi_+|^2 \sim \sin^2(\pi x / a)$$



$$|\psi_-|^2 \sim \cos^2(\pi x / a)$$



Lecture note 3.3
Homework 5.3

Sommerfeld vs. Bloch

	SOMMERFELD	BLOCH
QUANTUM NUMBERS (EXCLUDING SPIN)	$\mathbf{k}$ ($\hbar\mathbf{k}$ is the momentum.)	$\mathbf{k}, n$ ($\hbar\mathbf{k}$ is the crystal momentum and n is the band index.)
RANGE OF QUANTUM NUMBERS	$\mathbf{k}$ runs through all of k -space consistent with the Born-von Karman periodic boundary condition.	For each n , $\mathbf{k}$ runs through all wave vectors in a single primitive cell of the reciprocal lattice consistent with the Born-von Karman periodic boundary condition; n runs through an infinite set of discrete values.
ENERGY	$\mathcal{E}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m}$.	For a given band index n , $\mathcal{E}_n(\mathbf{k})$ has no simple explicit form. The only general property is periodicity in the reciprocal lattice: $\mathcal{E}_n(\mathbf{k} + \mathbf{K}) = \mathcal{E}_n(\mathbf{k})$.
VELOCITY	The mean velocity of an electron in a level with wave vector $\mathbf{k}$ is: $\mathbf{v} = \frac{\hbar\mathbf{k}}{m} = \frac{1}{\hbar} \frac{\partial \mathcal{E}}{\partial \mathbf{k}}$.	The mean velocity of an electron in a level with band index n and wave vector $\mathbf{k}$ is: $\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \frac{\partial \mathcal{E}_n(\mathbf{k})}{\partial \mathbf{k}}$.
WAVE FUNCTION	The wave function of an electron with wave vector $\mathbf{k}$ is: $\psi_{\mathbf{k}}(\mathbf{r}) = \frac{e^{i\mathbf{k} \cdot \mathbf{r}}}{V^{1/2}}$.	The wave function of an electron with band index n and wave vector $\mathbf{k}$ is: $\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k} \cdot \mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$ where the function $u_{n\mathbf{k}}$ has no simple explicit form. The only general property is periodicity in the direct lattice: $u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})$.

Semi-Classical Model

- Electrons are described as **wave packets (波包)**

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

- Uncertainty principle**

Ashcroft & Mermin, Chap. 12

- wave packets spread over many cells

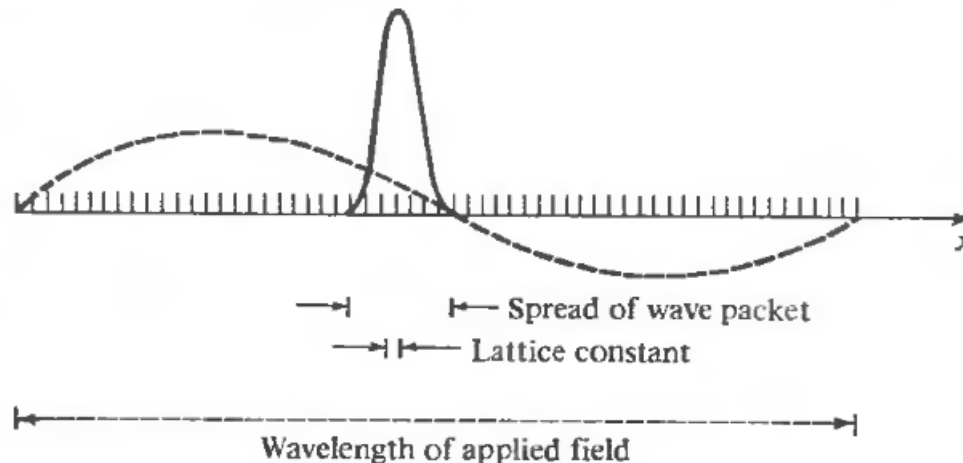
$$\Delta x \cdot \Delta p \geq \hbar / 2$$



$$\Delta x \cdot \Delta k \geq 1 / 2$$



$$\Delta x > na$$



Semi-Classical Model

- Electrons are described as **wave packets (波包)**

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

- **Uncertainty** principle
 - wave packets spread over many cells

$$\Delta x \cdot \Delta p \geq \hbar / 2 \quad \rightarrow \quad \Delta x \cdot \Delta k \geq 1 / 2 \quad \rightarrow \quad \Delta x > na$$

- External electric field varies slowly (DC or AC)

$$\lambda \gg a$$

Semi-Classical Model

- Electrons are described as **wave packets (波包)**

$$\psi(\mathbf{r}) = \exp(i\mathbf{k} \cdot \mathbf{r}) \cdot u_{\mathbf{k}}(\mathbf{r})$$

- **Collision / relaxation time τ**
 - originates from imperfect lattice (vibrations, impurities, defects, etc.), not from single atoms
 - mean free path

$$l = v_F \tau \gg a$$

- electrons in a perfect lattice at $T = 0$ K have no collision. (ideally, zero resistance, $\rho = 0$)

Semi-Classical Model

$$F = \frac{dp}{dt} = \hbar \frac{dk}{dt}$$

$$v = \frac{1}{\hbar} \frac{dE}{dk}$$



$$a = \frac{dv}{dt} = \frac{1}{\hbar} \frac{d^2 E}{dk \cdot dt}$$



$$\frac{F}{a} = \frac{\hbar^2}{\frac{d^2 E}{dk^2}} = m^*$$

effective mass

3D

$$\frac{1}{\hbar^2} \begin{pmatrix} \frac{\partial^2 E}{\partial k_x^2} & \frac{\partial^2 E}{\partial k_x \partial k_y} & \frac{\partial^2 E}{\partial k_x \partial k_z} \\ \frac{\partial^2 E}{\partial k_y \partial k_x} & \frac{\partial^2 E}{\partial k_y^2} & \frac{\partial^2 E}{\partial k_y \partial k_z} \\ \frac{\partial^2 E}{\partial k_z \partial k_x} & \frac{\partial^2 E}{\partial k_z \partial k_y} & \frac{\partial^2 E}{\partial k_z^2} \end{pmatrix} \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix} = \begin{pmatrix} \frac{dv_x}{dt} \\ \frac{dv_y}{dt} \\ \frac{dv_z}{dt} \end{pmatrix}$$

Effective Mass 有效质量

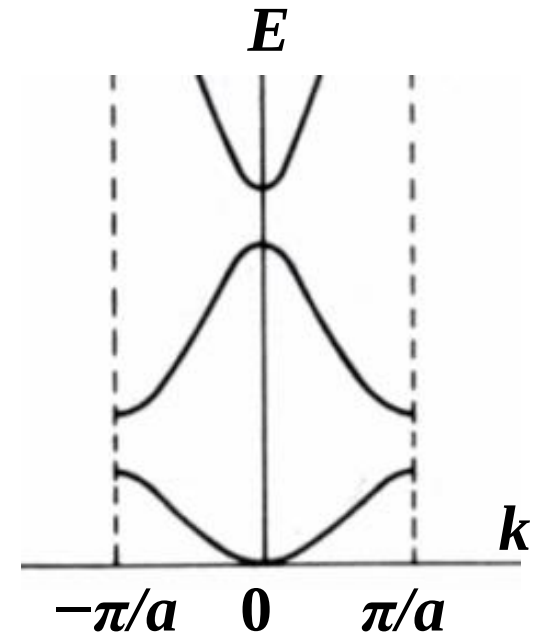
effective mass

$$\frac{1}{m^*} = \frac{1}{\hbar^2} \frac{d^2 E(k)}{dk^2}$$

The mass that an electron "seems" to have in a solid.
It has nothing to do with the free electron mass m_0

For 3D solids, a tensor form

$$\frac{1}{\mathbf{M}^*} = \frac{1}{\hbar^2} \begin{pmatrix} \frac{\partial^2 E}{\partial k_x^2} & \frac{\partial^2 E}{\partial k_x \partial k_y} & \frac{\partial^2 E}{\partial k_x \partial k_z} \\ \frac{\partial^2 E}{\partial k_y \partial k_x} & \frac{\partial^2 E}{\partial k_y^2} & \frac{\partial^2 E}{\partial k_y \partial k_z} \\ \frac{\partial^2 E}{\partial k_z \partial k_x} & \frac{\partial^2 E}{\partial k_z \partial k_y} & \frac{\partial^2 E}{\partial k_z^2} \end{pmatrix}$$



m^* is a function of k ,
can be smaller or
larger than m_0 , even
can be negative

$$m_0 = 9.11 \times 10^{-31} \text{ kg}$$

Effective Mass 有效质量

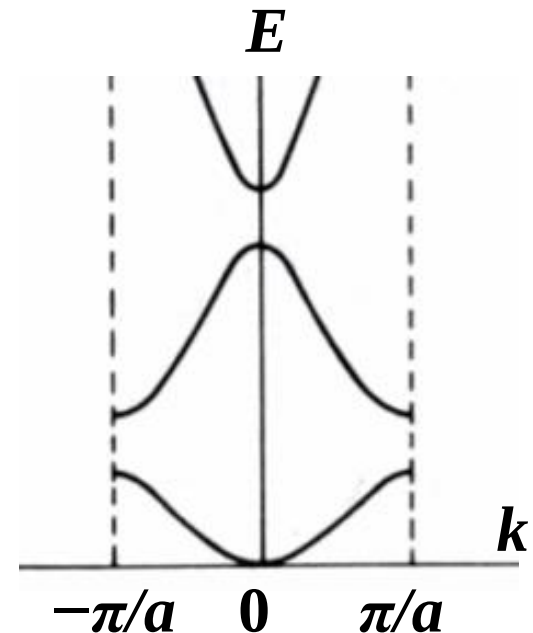
effective mass

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For 3D solids, a tensor form

$$\frac{1}{\mathbf{M}_{ij}^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k_i \partial k_j}$$



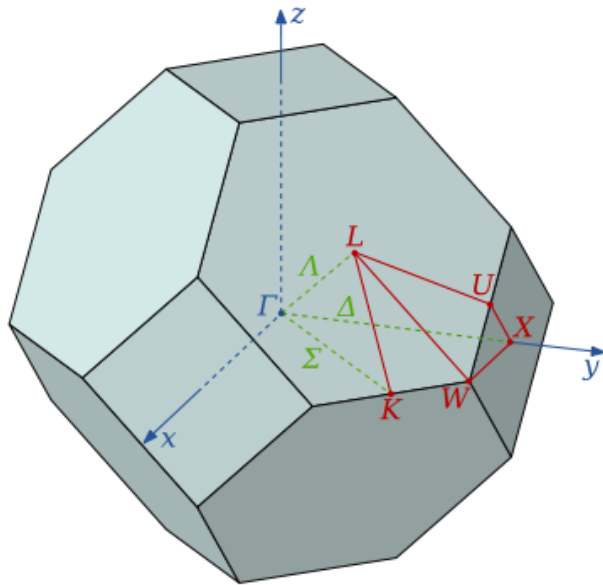
**m^* is a function of k ,
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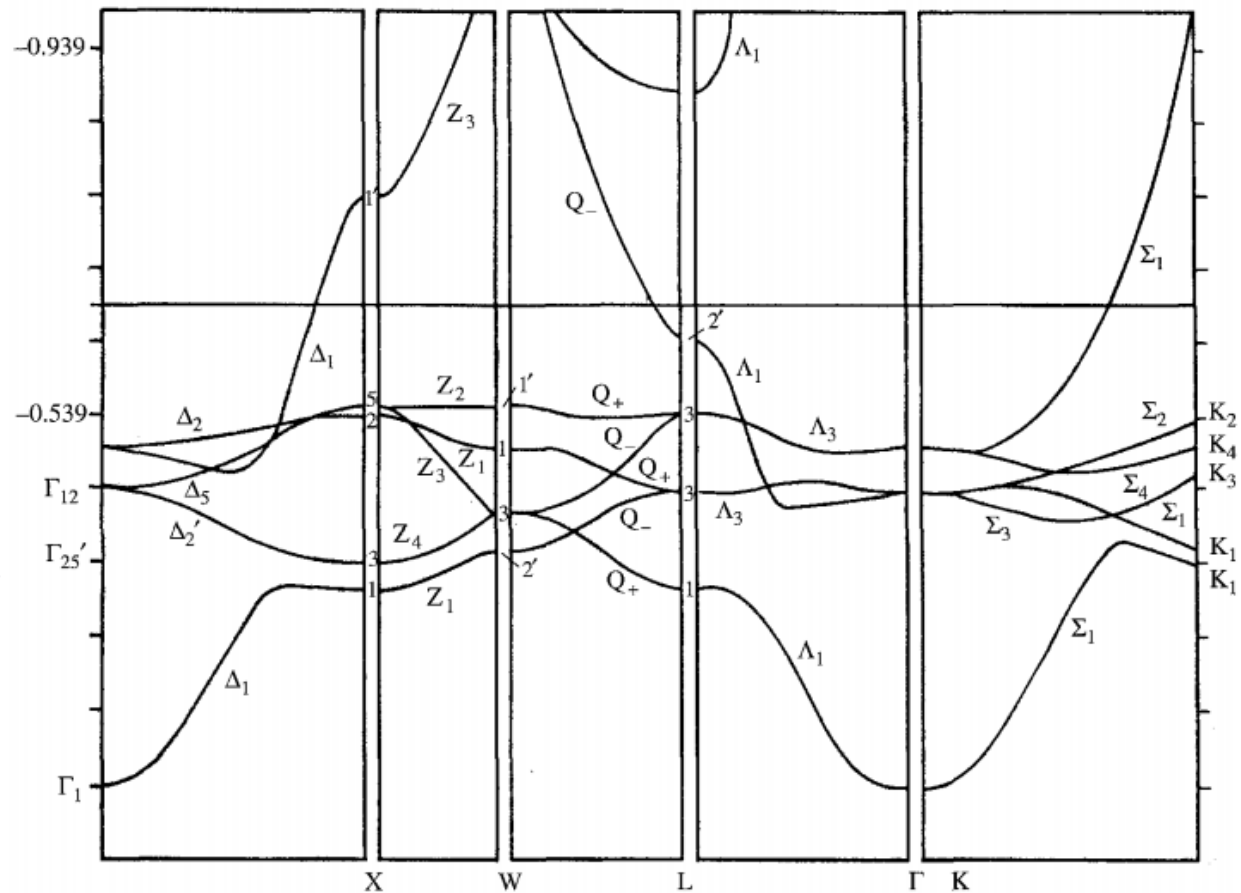
Band Structure in 3D

copper (FCC)

$$E = E(k_x, k_y, k_z)$$



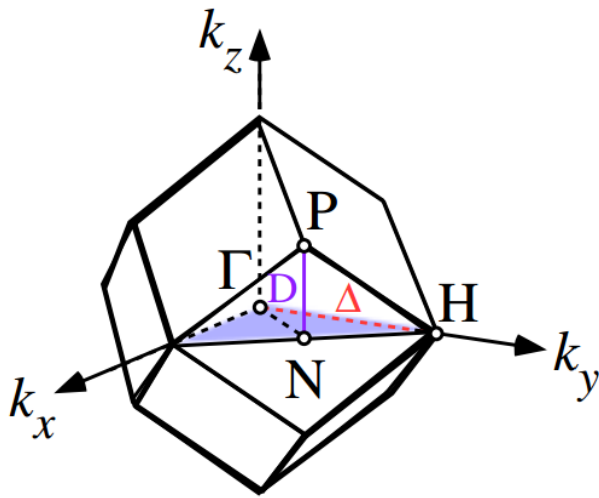
First BZ



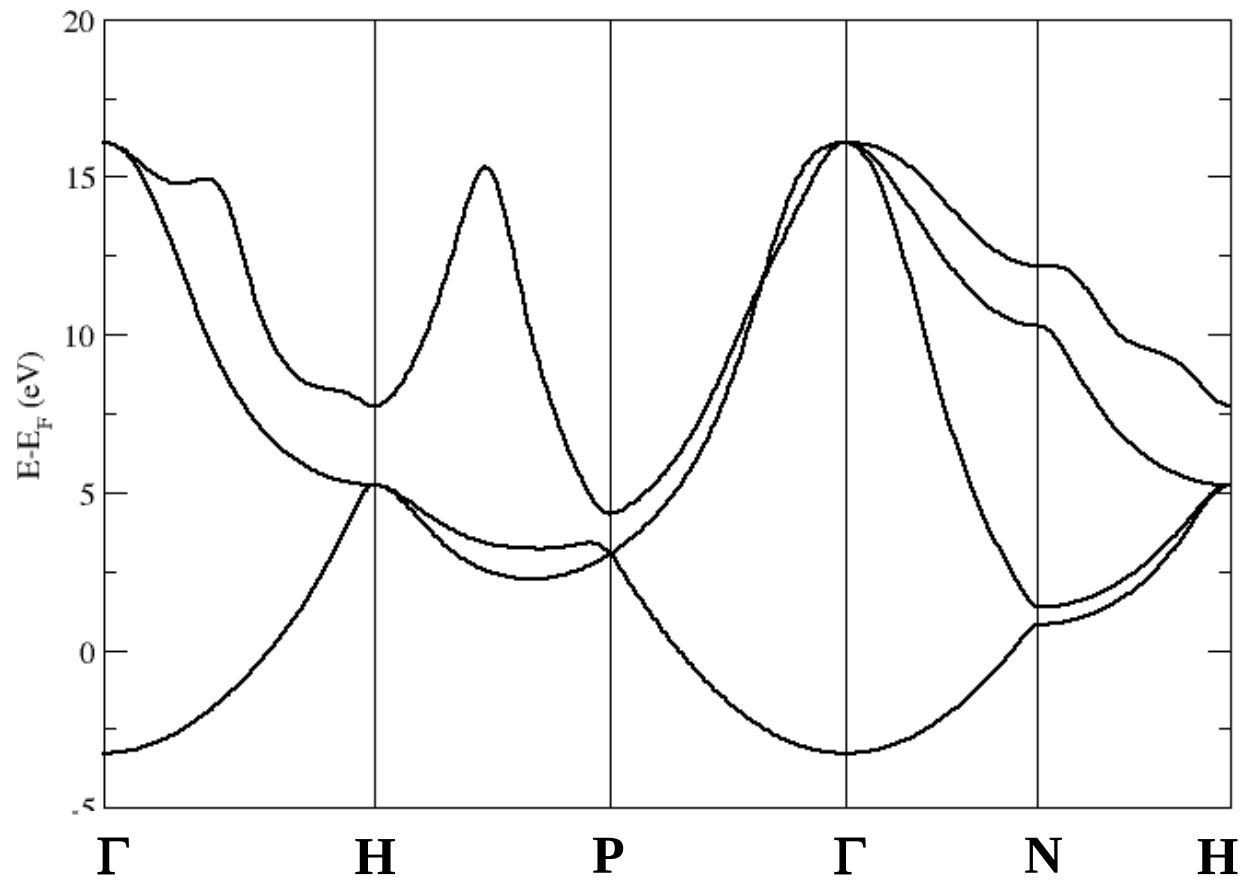
Band Structure in 3D

sodium (BCC)

$$E = E(k_x, k_y, k_z)$$



First BZ

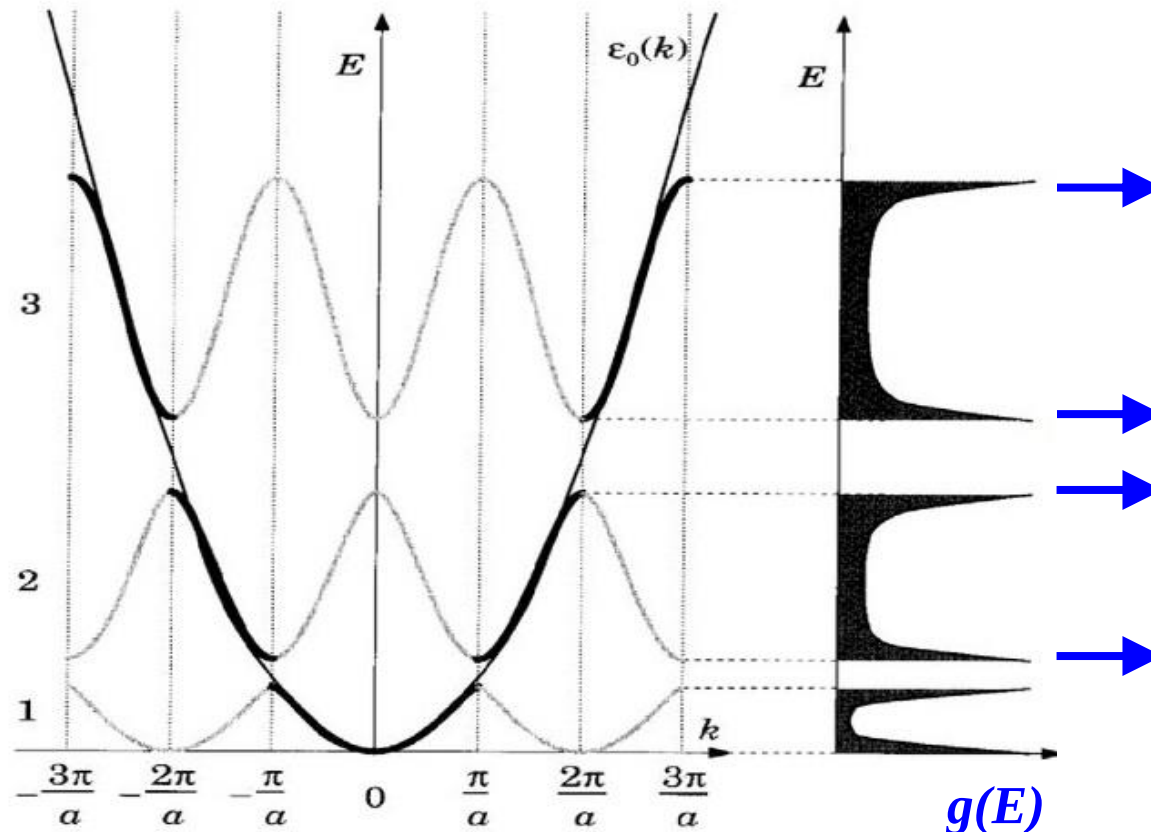


Density of States (DOS) 态密度

$$g(E) = \frac{dn}{dE}$$

DOS - number of energy states/levels per unit energy in $[E, E+dE]$, per unit volume

1D



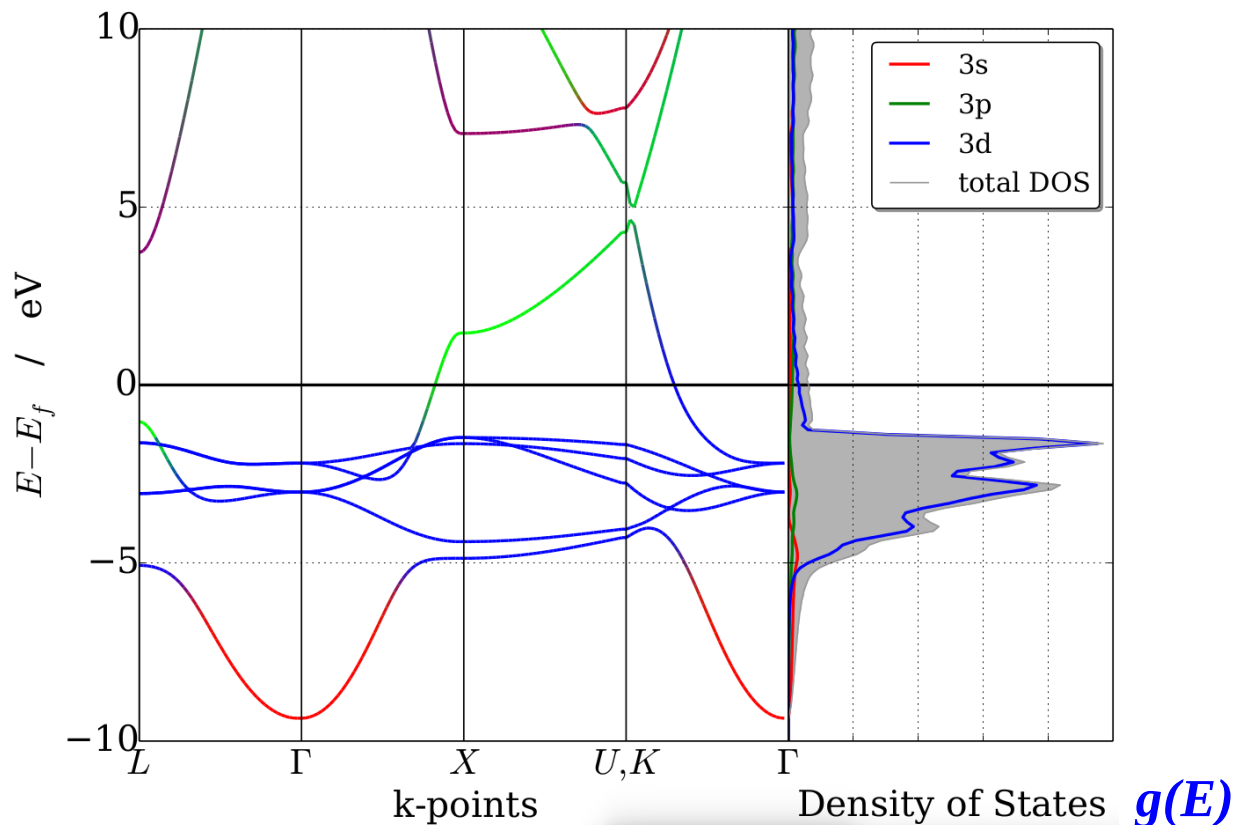
van Hove singularities

Density of States (DOS) 态密度

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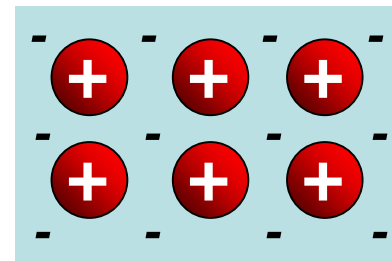
DOS - number of energy states/levels per unit energy in $[E, E+dE]$, per unit volume

**3D
copper**



Bloch Model - Review

- Electrons are *not free*, but *in a periodic potential* formed by the atomic lattice



positive ions
+
electron cloud

- Born-Oppenheimer Approximation
 - ▣ The behaviors of electrons and nuclei can be calculated separately.
- Independent Electron Approximation
 - ▣ We still assume electrons are independent and do not interact with each other

Real Electrons in Solids are Nightmares

$$\hat{H}\Psi = E\Psi$$

many-body problem '多体'问题

$$\sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \nabla_i^2 \Psi - Ze^2 \sum_{\mathbf{R}} \frac{1}{|\mathbf{r}_i - \mathbf{R}|} \Psi \right) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} \Psi = E\Psi$$

free electron

electron-atom
interaction

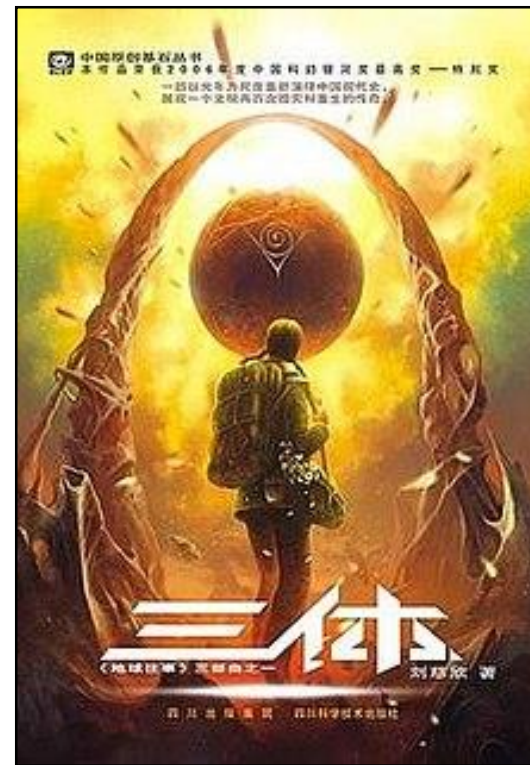
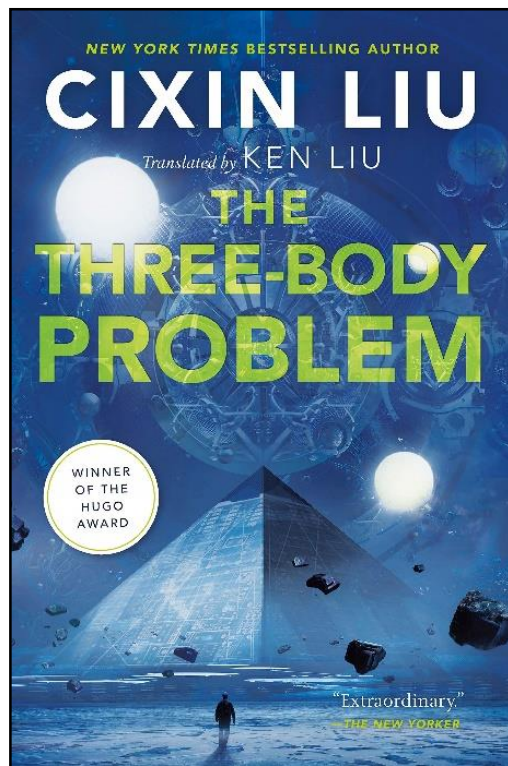
electron-electron
interaction

Wave function is complicated and impossible to solve

Real Electrons in Solids are Nightmares

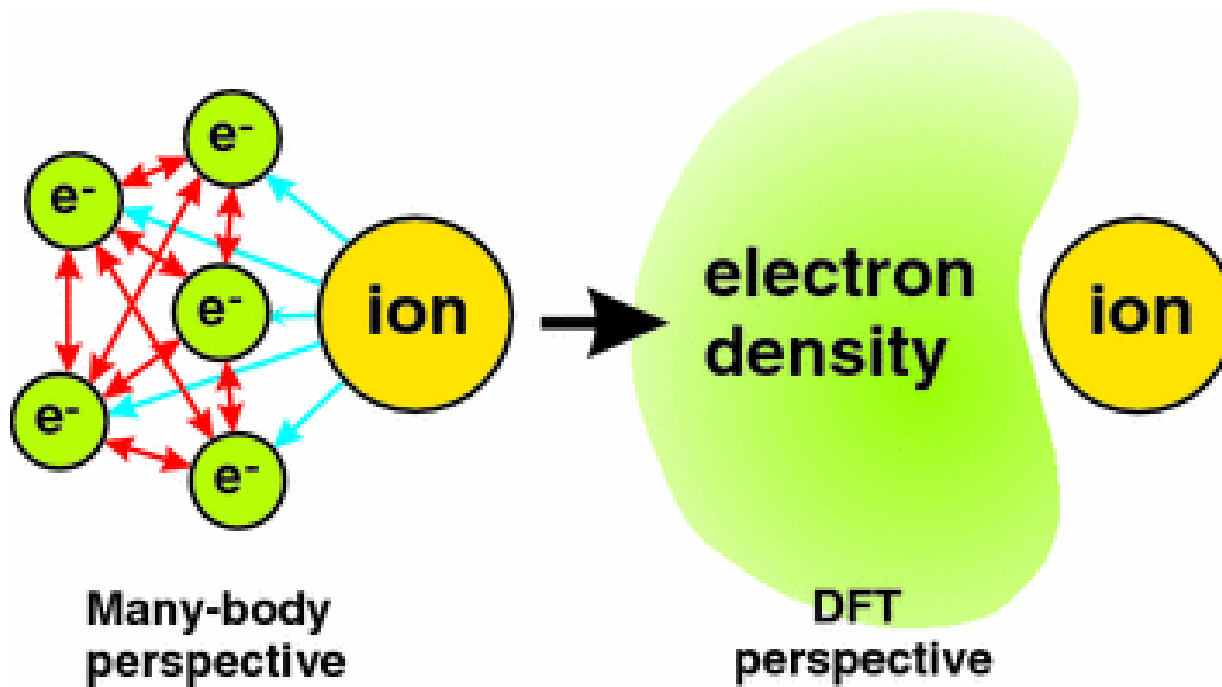
$$\hat{H}\Psi = E\Psi$$

many-body problem '多体'问题



Wave function is complicated and impossible to solve

Density Functional Theory (DFT) 密度泛函理论



wave function $\rightarrow$ electron density function

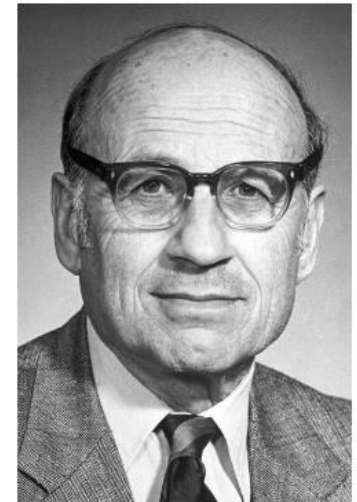


Photo from the Nobel Foundation archive.

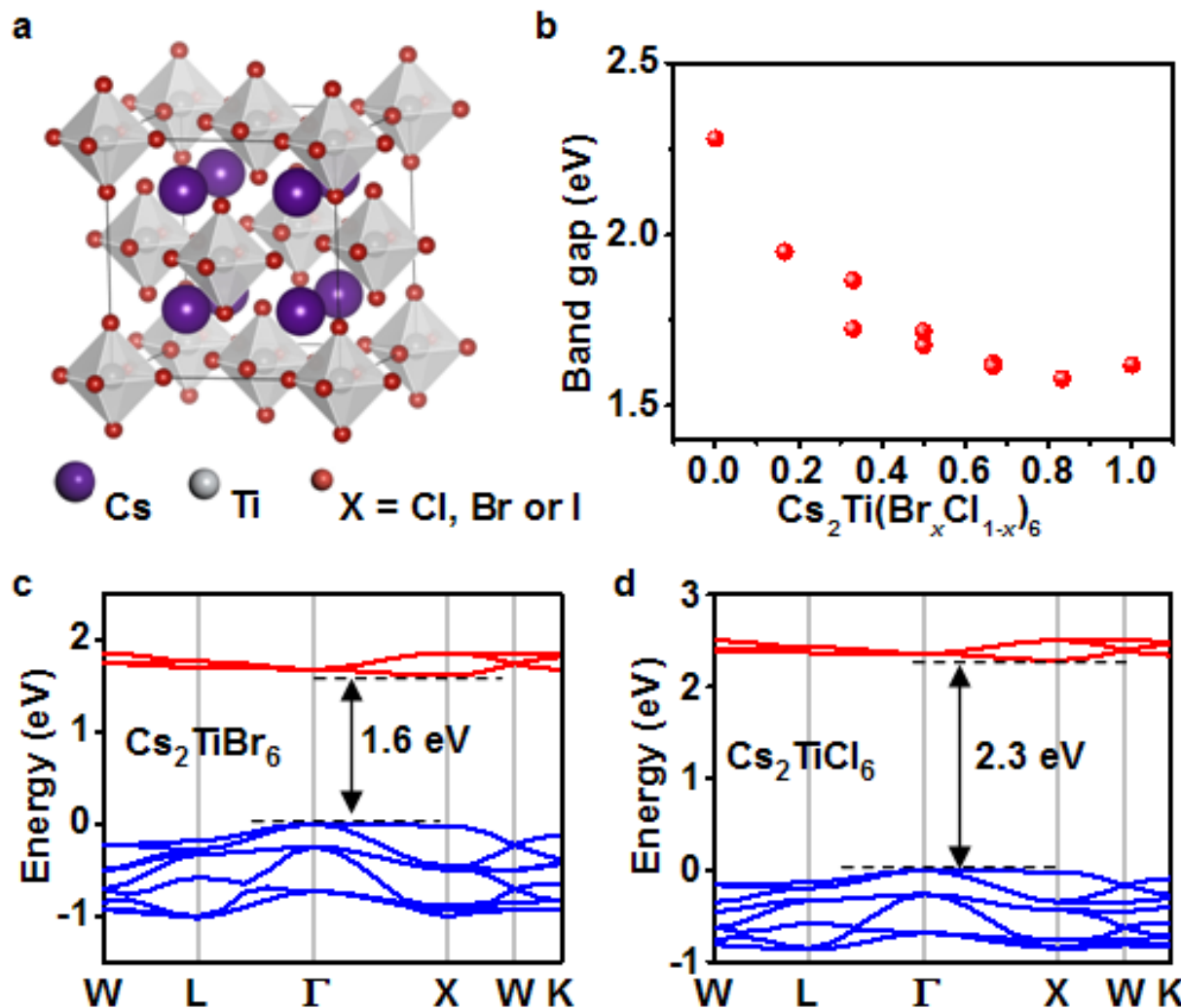
Walter Kohn

Prize share: 1/2

Nobel Prize in Chemistry

1998

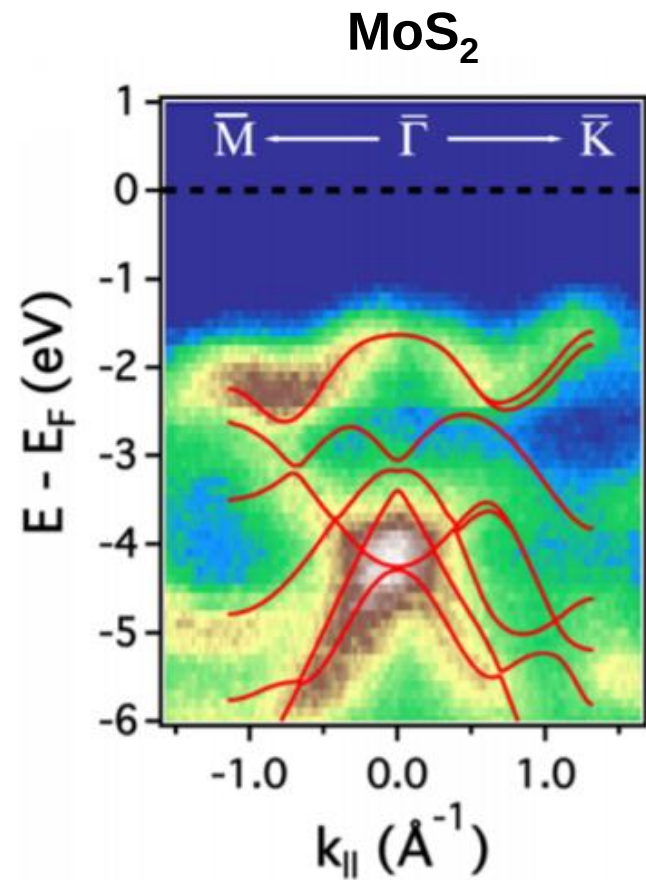
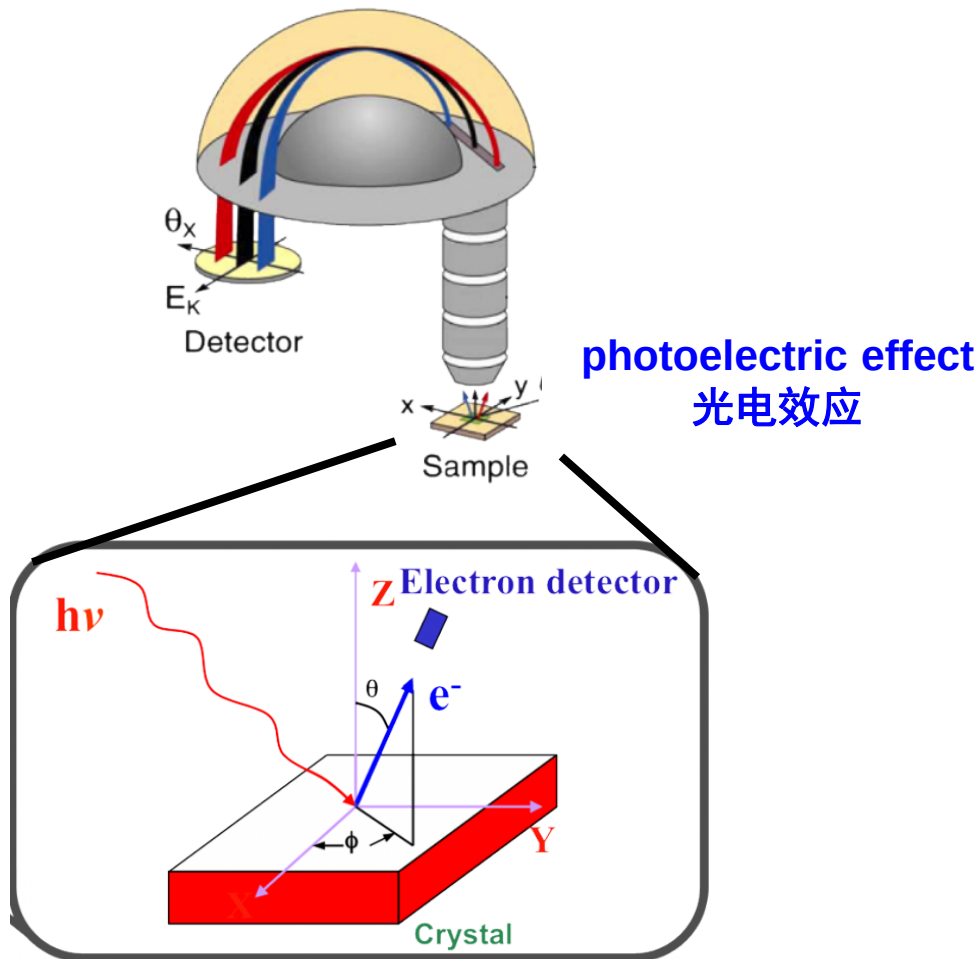
Band Structures by DFT - Example



work by
2016级本科生
成大立

Measurement of Band Structures

Angle-Resolved Photoemission Spectroscopy (ARPES)



W. Jin, et al, *Phys. Rev. Lett.* **111**, 106801 (2013)

Next Class

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Thank you for your attention